

MUKHAMAD KHAFID MAASSOBIRIN

Artificial Intelligence Engineer

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PROFESSIONAL SUMMARY

Data Science graduate with hands-on research experience designing and deploying Retrieval-Augmented Generation (RAG) systems, including a hierarchical RAG architecture evaluated across four benchmark datasets with rigorous statistical testing. Experienced building end-to-end AI pipelines, from embedding and retrieval design to LLM serving and deployment, with additional hands-on exposure to computer vision through object detection systems built during internship experience. Complemented by working knowledge of data engineering and data analysis, with a continuous drive to deepen technical expertise across emerging AI, ML and data tools.

EXPERIENCE

Data Scientist Intern - PT Lautan Natural Krimerindo

February 2025 - June 2025

- Built an internal RAG-based chatbot using Python, Gemini 1.5 Flash-Lite, and Qdrant, with a Random Forest query-routing classifier to direct employee questions to the relevant document category covering company regulations, organization structure, and company profile, falling back to general LLM responses for out-of-scope queries.
- Built an end-to-end planogram compliance pipeline using YOLO, Python, and BigQuery, training an object detection model on labeled retail shelf images of company and competitor products, then validating shelf placement against internal rules such as correct product clustering and size-based ordering, eliminating the need for manual review of shelf audit photos for the sales team.
- Developed a Google Maps scraping pipeline using Python and Selenium to extract business details, contact info, ratings, and facility keywords from 10,850 coffee shop locations, applying anti-detection techniques to avoid Google's bot protection, to support the sales team in prioritizing high-potential outlets.
- Built a conversational practice tool combining speech-to-text, an LLM, and text-to-speech to help employees improve fluency in foreign languages such as English and Mandarin ahead of meetings with overseas clients.

Data Analyst Intern - BPBD Kota Surabaya

January 2025 - February 2025

- Cleaned, validated, and consolidated daily accident and emergency incident logs into a structured database to ensure data integrity.
- Designed and built an interactive Excel dashboard utilizing advanced formulas and automated charts to track and visualize accident trends.
- Conducted exploratory data analysis (EDA) on historical accident data to identify high-risk locations and peak times, providing actionable insights for academic reporting and organizational review.

PROJECTS

Hierarchical IRCoT RAG Chatbot

- Designed a hierarchical RAG architecture combining RAPTOR-style tree construction with IRCoT-guided reasoning traversal to answer questions requiring deep reasoning beyond semantic similarity, especially on long and complex documents.
- Built the embedding and clustering pipeline using multilingual-e5-large-instruct, UMAP for dimensionality reduction, and BIC-optimized Gaussian Mixture Models to structure documents into a hierarchical tree.
- Implemented an adaptive DFS traversal algorithm with dual-threshold pruning and interleaved reasoning generation to guide retrieval toward the most relevant nodes.
- Evaluated the system across four benchmark datasets using statistical significance testing, showing consistent gains on NarrativeQA and QASPER while identifying trade-offs on datasets requiring holistic or short-context understanding.
- Deployed the system with a FastAPI backend, using Qdrant as the vector store and vLLM to serve the Qwen2.5-7B-Instruct-AWQ model, and published the project on GitHub.

Premier League Data Warehouse

- Built an automated data pipeline that extracts player, team, fixture, and gameweek statistics from the official Fantasy Premier League API on a scheduled basis using GitHub Actions.
- Loaded raw data into BigQuery and used dbt to transform it into structured, modeled tables for downstream analysis.
- Designed the pipeline to run on a recurring schedule without manual intervention, simulating a production-style data workflow.

EDUCATION

Bachelor of Science in Data Science - Universitas Pembangunan Nasional “Veteran” Jawa Timur

July 2022 - July 2026

- GPA: 3.79 / 4.00
- Thesis: Tree Traversal Optimization through Interleaved Chain-of-Thought in Hierarchical Retrieval-Augmented Generation Architecture

TECHNICAL SKILLS

Programming Languages:

- Python, SQL

Data Engineering:

- ETL/ELT Pipelines, Google BigQuery, dbt, Vector Databases (Qdrant), Web Scraping (Selenium), REST API Integration (Postman), Apache Airflow, PostgreSQL

Artificial Intelligence & Machine Learning:

- Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Scikit-learn, PyTorch, Computer Vision (YOLO)

Data Analysis:

- Exploratory Data Analysis (EDA), Microsoft Excel, Looker Studio

Tools & Platform:

- Git, GitHub, GitHub Actions, FastAPI, Jupyter Notebook, VS Code

CERTIFICATIONS

- TOEFL ITP - Score: 460
2026
- Junior Web Programmer - BNSP (Badan Nasional Sertifikasi Profesi)
2025

PUBLICATIONS

- Improving Tree Traversal in Hierarchical RAG via Reasoning-Aware Approach with Interleaving Chain-of-Thought
Bit-Tech (Binary Digital - Technology), Sinta 4 (Accepted, forthcoming August 2026)
- Analisis Komparatif Model Abstractive Summarization dalam Pembentukan Struktur Hierarkis Dokumen pada Arsitektur RAPTOR
Journal of Data, Engineering, and Science of Health, Vol. 1 No. 1 (2026)

ADDITIONAL INFORMATION

- Languages: Indonesian (Native), English (Professional Working Proficiency)
- Organizations & Leadership:
 - HIMASADA (Data Science Student Association) - Staff of Cadre Mentoring Division (2023 - 2024)
 - Data Science Orientation Program - Chairman of Department Orientation Committee (2023 - 2024)
 - Himpunan Mahasiswa Islam (HMI) - Chairman of PTKP Department (2025 - 2026)
 - Himpunan Mahasiswa Islam (HMI) - Project Leader of LK 1 Training (2024)